COURSE INFORMATION SHEET

Course Code: CS24201
Course Title: Data Structure and Algorithms
Pre-requisite(s): CS24101 Programming for Problem Solving
Co- requisite(s): MA24205 Discrete Mathematics
Credits: L: 3 T: 0 P: 0
Class schedule per week:
Class:
Semester / Level: III
Branch: CSE
Name of Teacher:

COURSE OBJECTIVES

This course envisions to impart to students to:

	To be familiar with basic data structures and algorithm analysis.
	To understand basic concepts about arrays, linked lists .
	To understand basic concepts about stack, queue.
	To analyze concepts of searching and sorting techniques.
	To understand concepts of non-linear data structures.

COURSE OUTCOMES (COs)

After the completion of this course, students will be able to:

CO1	Be conversant with basic data structures and algorithm analysis.
CO2	Be able to apply the concepts of arrays, linked lists .
CO3	Be able to apply stack and queue data structures in solving problems.
CO4	Be able to analyze searching and sorting algorithms.
CO5	Be able to use non-linear data structures in solving problems.

SYLLABUS

MODULE	(NO. OF LECTURE HOURS)
Module – I **Basics of Data Structures and Algorithm, Time and Space Complexity, Asymptotic Notations, Time complexity analysis of non-recursive and recursive algorithms, Examples of multi-dimensional array, polynomial operations and sparse matrix**	**5**
Module – II **Singly Linked List: concept, representation and operations, Circular Linked List, Doubly Linked List, Multi-ply linked list and their Applications**	**8**
Module – III **Stack: basic operations using Array and LL, Queue: basic operations using Array and LL, Circular Queue and Variants of Queue, Stack and Queue**	**10**

Applications	
Module – IV **Binary Search, Hash Search, Selection Sort, Insertion Sort, Quick Sort, Merge Sort, Radix Sort, External Sorting: k-way merging approach, Analysis of Search and Sorting Algorithms**	**7**
Module – V **Basic concepts and terminologies of Binary Search Tree, Height Balanced Trees and Heap, Disjoint Set, Graph: concept and terminologies, Concept of Breadth First Search, Depth First Search, Spanning Tree.**	**10**

TEXTBOOKS:
Sahni Horwitz,, Freed Anderson, Fundamentals of Data Structures in C, 2 nd Edition (or latest) , University Press.(T1)

REFERENCE BOOKS:
1. Thareja Reema, Data Structures Using C, 2nd Edition, Oxford University Press.(R1)
2. Tanenbaum, Langsam, Augenstein, Data Structures using C, Pearson. (R2)

GAPS IN THE SYLLABUS (TO MEET INDUSTRY/PROFESSION REQUIREMENTS)

POS MET THROUGH GAPS IN THE SYLLABUS

TOPICS BEYOND SYLLABUS/ADVANCED TOPICS/DESIGN

POS MET THROUGH TOPICS BEYOND SYLLABUS/ADVANCED TOPICS/DESIGN

COURSE OUTCOME (CO) ATTAINMENT ASSESSMENT TOOLS & EVALUATION PROCEDURE

DIRECT ASSESSMENT

Assessment Tool	% Contribution during CO Assessment
Continuous Internal Assessment	50
Semester End Examination	50

Continuous Internal Assessment	% Distribution
Mid semester examination	25
Quiz1 & Quiz2/Assignment/Seminar Presentation	20
Teacher's Assessment	5

Assessment Components	CO1	CO2	CO3	CO4	CO5
Continuous Internal Assessment	20%[10]	20%[10]	20%[10]	20%[10]	20%[10]
Semester End Examination	20%[10]	20%[10]	20%[10]	20%[10]	20%[10]

INDIRECT ASSESSMENT
11. Student Feedback on Course Outcome

COURSE DELIVERY METHODS

CD1	Lecture by use of boards/LCD projectors
CD2	Tutorials/Assignments
CD3	Seminars/ Quiz (s)
CD4	Mini projects/Projects
CD5	Laboratory experiments/teaching aids
CD6	Industrial/guest lectures
CD7	Self-Learning, Group Study, Coding Contest

MAPPING BETWEEN COURSE OUTCOMES AND POs and PSOs

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO 2	PS O3
CO 1	**3**	**3**	**3**	**3**	**3**	**2**	**0**	**0**	**1**	**2**	**2**	**2**	**3**	**3**	**2**
CO 2	**3**	**3**	**3**	**3**	**3**	**2**	**0**	**0**	**1**	**2**	**2**	**2**	**3**	**3**	**2**
CO 3	**3**	**3**	**3**	**3**	**3**	**2**	**0**	**0**	**1**	**2**	**2**	**2**	**3**	**3**	**3**
CO 4	**3**	**3**	**3**	**3**	**3**	**2**	**0**	**0**	**1**	**2**	**2**	**2**	**2**	**3**	**2**
CO 5	**3**	**3**	**3**	**3**	**3**	**2**	**0**	**0**	**1**	**2**	**2**	**2**	**2**	**3**	**3**

Grading: No correlation – 0, Low correlation - 1, Moderate correlation – 2, High Correlation - 3

MAPPING BETWEEN COURSE OUTCOMES AND COURSE DELIVERY METHOD

Course Outcomes	Course Delivery Method
CO1	CD1, CD2, CD3
CO2	CD3, CD5
CO3	CD3, CD5, CD7
CO4	CD2, CD3, CD4, CD6, CD7
CO5	CD1, CD3, CD5, CD7